

# INSTITUT FÜR SPORTBODENTECHNIK

Dipl.-Phys. B. Härting VDI



## TEST CERTIFICATE

Test according to EN 14904:2006

1<sup>st</sup> issue

Applicant:

Systembau Horst Babinsky GmbH  
Fabrikstraße 14  
D-83371 Stein

and

KINON Porz GmbH  
Concordiaplatz 2  
D-51143 Köln

Subject:

Area-elastic sports floor  
„ASB-SGG-Sportsfloor“

Tested in terms of EN 14904:2006

Date of the report:

January 11, 2010

Test:

L 3853

Page 1 of

5 text pages

Attachment:

3 pages

Without approval by the IST any reproduction is allowed in full only.



DIPL.-PHYS. B. HÄRTING VDI

Test certificate L3853 of January 11, 2010  
Page 2 of 5

Systembau Horst Babinsky GmbH  
Fabrikstraße 14  
D-83371 Stein

and

KINON Porz GmbH  
Concordiaplatz 2  
D-51143 Köln

Markkleeberg, January 11, 2010-04-27

## TEST CERTIFICATE

|                              |                                                      |
|------------------------------|------------------------------------------------------|
| IST ident. mark:             | L 3853/Hä                                            |
| Subject:                     | Indoor sports flooring<br>Test acc. to EN 14904:2006 |
| Designation of floor tested: | Area-elastic sports floor<br>„ASB-SGG-Sportsfloor“   |
| Place of test:               | Test chamber of Systembau Horst Babinsky GmbH        |
| Date of test:                | June 3, 2009                                         |
| Number of test samples:      | 1 sample floor                                       |
| Size of test samples:        | 3.20 m x 4,90 m                                      |
| Test sample set up:          | 3 <sup>rd</sup> June 2009                            |
| Sampling:                    | 2 elastic strips 20 mm x 50 mm x 500 mm              |

## 1. FLOOR STRUCTURE

The construction of the substructure met the requirements stated in Attachment A3 of DIN V 18032-2:2001-04 for area-elastic sports flooring with elastic structure with crosswise one over the other laid springboards, here flat aluminium, and without a sub floor layer. Axis distances were 516 mm and 532 mm (lower layer 50 mm x 8 mm), resp. 477 mm and 487 mm (upper layer 50 mm x 8 mm). Elevation comprised no elastic elements.

Elastic strips (20 x 50 mm) were positioned between the 4 set up glass panels SSG STAPID 55.4 made of 1 x 5 mm SGG SERALIT P 309 on cauterized surface, 1.52 mm PVB foil and 1 x 5 mm SECURIT and the substructure. All necessary parameters are available from IST L.

Size of tested floor was about 3.20 x 4.90 m<sup>2</sup>.

## 2. TESTING

According to EN 14904:2006 the floor was tested in the chamber at the premises of Systembau Horst Babinsky GmbH using the "Artificial Athlete 95" ("Künstliche Sportler 95").

The test climate applied was 23/50-2 as per EN 14904:2006.

The measuring points were arranged as given in Attachment A.6 of EN 14904:2006 for area-elastic sports flooring with an elastic structure.

### 3. TEST RESULTS

#### 3.1. Test of total system

In the following table the measuring results are summarised and compared with the requirements of EN 14904:2006. The individual measured values can be gathered from Attachment 1, the position of the system measuring points from Attachment 2. The test results relate to the test sample only which was arranged as described (floor sample).

| Test acc. to<br>EN 14904:2006                              | Measuring results<br>min.    Aver-    max.<br>value    age    value |     |     | Requirements as per<br>EN 14904:2006<br>area-elastic flooring |
|------------------------------------------------------------|---------------------------------------------------------------------|-----|-----|---------------------------------------------------------------|
| Shock resorption <sup>ab</sup><br>FR <sub>55</sub> in    % | 60                                                                  | 65  | 70  | min. 25 max. 75<br>Deviation: +/- 5% from average             |
| Standard deformation<br>StD    in    mm                    | 3.4                                                                 | 3.9 | 4,6 | max. 5.0                                                      |
| Rolling load without<br>damage    in    N                  | 1 500<br>no visible damage                                          |     |     | ≥ 1 500<br>no visible damage                                  |
| Ball rebound<br>BR    in    %                              | 91                                                                  | 93  | 95  | min. 90<br>Deviation: +/- 3% from average                     |

### 3.2 Technical data of top layer coating

(Manufacturer information from product specifications records)

3.2.1 SGG STADIP 55.4 made of 1 x 5 mm SGG SERALIT P 309 on cauterized surface, 1.52 mm PVB foil and 1 x 5 mm SECURIT

| Feature             | Analysis    | characteristic  |
|---------------------|-------------|-----------------|
| Fire behaviour      | EN 13501-1  | F <sub>fl</sub> |
| Sliding friction *) | DIN 18032-2 | 0.52 – 0.57     |
| Light reflection    | DIN 5036-1  | not reported    |

\*) according to test report 55150/900 9943-1/Kf/C of 30 August, 2005

## 4. ASSESSMENT

The assessment is based on the limits specified of EN 14904:2006.

These requirements have been met by the tested sports floor at all measuring point.

With respect to the results of force reduction and standard deformation as per Attachment B tables B1 and B2 column A, the sports floor is classified Type A 4.

## Individual measured values

| Test                                      | Measuring points  |                   |                   | Average |
|-------------------------------------------|-------------------|-------------------|-------------------|---------|
|                                           | 1                 | 2                 | 3                 |         |
|                                           | 4                 | 5                 | 6                 |         |
|                                           | 7                 | 8                 | 9                 |         |
| Shock absorption<br>FR <sub>55</sub> in % | 62<br>69<br>61    | 67<br>67<br>69    | 60<br>64<br>70    | 65      |
| Standard deformation<br>StD in mm         | 3.5<br>4.0<br>3.9 | 3.7<br>4.3<br>4.8 | 3.4<br>3.8<br>4.6 | 3.9     |
| Ball rebound<br>BR in %                   | 94<br>94<br>94    | 93<br>92<br>92    | 95<br>91<br>92    | 93      |
| Friction *)                               | 0.54<br>0.59      | 0.52<br>0.56      | 0.57              | 0.57    |

\*) according to test report 55150/900 9943-1/Kf/C of 30 August, 2005

## **Arrangement of measuring points:**

- 1 over an elastic pad
- 2 over an intersection point upper / lower layer
- 3 at a joint of upper layer
- 4 at a joint of lower layer
- 5 between two elastic pads on the field
- 6 centrically on the upper layer on the field
- 7 at a joint of the longer side of glass panels
- 8 at a joint of the shorter side of glass panels above an elastic pad
- 9 at a cross joint of glass panels above an intersection point



## Grundrissplan

### Testboden-Unterkonstruktion

Obergurt Flachalu 50x8 mm

Untergurt Flachalu 50x8 mm

Schloß Stein - 2009

